CRISTOFER MONTES LOZADA

Mexico City, Mexico
Email: hello@cmontes.dev
LinkedIn: linkedin.com/in/cristofer-montes-lozada/
Web Page: cmontes.dev/

PROFESSIONAL SUMMARY

Backend-oriented software developer with strong foundations in data structures, algorithms, and system design. Experience building REST APIs and data-driven applications using Java and Spring Boot, with a focus on performance, scalability, and maintainable software architecture.

TECHNICAL SKILLS

Programming Languages: Proficient: Java; Beginner: C#, Python, JavaScript
Backend Development: Spring Boot, Spring Data JPA (Hibernate), REST API design, object-oriented design, layered architecture
Databases: SQL (PostgreSQL, MySQL, Oracle), database design, normalization, query optimization (indexing)
API & Data Handling: JSON processing, data serialization/deserialization, API documentation (Swagger/OpenAPI)
Frontend web technologies: HTML5, CSS3
Software Engineering: Data structures, algorithms, complexity analysis (Big O), error handling, validation
Version Control & Collaboration: Git (branching strategies, Git Flow, pull requests, code reviews, conventional commits), GitHub, GitLab
AI-Assisted Development: GitHub Copilot, OpenAI Codex (code generation, debugging, refactoring)
Tools: Postman, IntelliJ IDEA, PyCharm, VS Code, Visual Studio Community

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PROFESSIONAL EXPERIENCE

Backend Developer Intern

badak | Mexico City | September 2025 – April 2026

- Proposed technical solutions to the team tech lead and prevented potential bugs
- Modeled data using a CASE tool
- Developed REST APIs using Java and Spring Boot
- Implemented data processing logic and global error handling
- Worked with SQL databases and ORMs for data persistence
- Debugged and troubleshot backend applications using the IntelliJ IDEA debugger, log analysis, browser developer tools, and SQL
- Assisted my training colleagues in developing their services through pair programming sessions and Git mentoring

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EDUCATION

Bachelor of Science in Computer Systems Engineering
Instituto Politécnico Nacional, Mexico City

Completed coursework, degree pending
Relevant Coursework: Data Structures and Algorithms, Algorithm Analysis and Design, Object-Oriented Programming, Databases, Software Engineering

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ACADEMIC PROJECTS

Electronic Voting System Prototype
Terminal Project | 2024

- Implemented Shamir's Secret Sharing for distributed key management
- Designed an algorithm to verify correctness of vote anonymization (mixing)
- Developed a Python module for vote validation, decryption, and counting
- Contributed to system architecture and security design
- Applied system isolation configurations

Tools & technologies: Python, cryptography, algorithm design

Set Algebra Learning System (Venn Diagrams)
Social Service | 2024

- Developed a desktop application with role-based access (student and teacher)
- Implemented encrypted data serialization
- Built an analytics module to rank exercises by error rate
- Integrated email-based authentication using SMTP
- Implemented search and filtering by student ID

Tools & technologies: C#, .NET, SMTP, data processing

Monitoring System with Microcontroller Integration
Instrumentation and Control | 2023

- Programmed an ATMega8535 microcontroller for sensor data processing
- Developed a C# desktop application for real-time data visualization
- Implemented serial communication between hardware and software
- Performed simulation and physical testing

Tools & technologies: C, C#, microcontrollers, serial communication

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LANGUAGES

Spanish: Native
English: Upper-Intermediate (B2 CEFR)